

Transformer Architecture Overview

The Transformer architecture, introduced in the 2017 paper 'Attention Is All You Need', replaced recurrence and convolutions with a mechanism called self-attention.

Self-attention allows each token in a sequence to weigh the importance of every other token when building its representation, which enables parallel processing across the entire sequence instead of the step-by-step processing required by RNNs.

A standard Transformer has an encoder stack and a decoder stack. The encoder maps an input sequence to a sequence of continuous representations, while the decoder generates an output sequence one token at a time, attending both to its own previous outputs and to the encoder's representations.

GPT-style models use a decoder-only architecture: they drop the encoder entirely and use masked self-attention so each token can only attend to previous tokens, which makes the model naturally suited to autoregressive text generation.

BERT, in contrast, uses only the encoder stack with bidirectional self-attention, allowing every token to attend to every other token in the sequence. This makes BERT strong at understanding tasks like classification, but it is not designed for free-form text generation the way GPT is.

Positional encodings are added to token embeddings because self-attention itself has no built-in notion of token order. The original paper used fixed sinusoidal functions, while many later models use learned positional embeddings instead.

Multi-head attention runs several attention operations in parallel, each with its own learned projection, allowing the model to jointly attend to information from different representation subspaces at different positions.